

SHASHANK KATIYAR

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EDUCATION

Carnegie Mellon University

Master of Science in Computational Biology

GPA: 4.10/4.33

Pittsburgh, PA

December 2024

Indian Institute of Technology Kanpur

Bachelor of Technology in BioSciences and Bioengineering

GPA: 9.1/10 – Department Rank: 1

Kanpur, India

June 2023

SKILLS

Programming: Python, R, C/C++, Golang, Shell Scripting (UNIX/Linux)

ML Techniques: Random Forest, Gradient Boosting, CNN, Reinforcement Learning, Transformers

Bioinformatics: Single Cell Data Analysis (Scanpy, Seurat), Spatial Data Analysis (spatialdata, squidpy)

Toolkit: Pytorch, Pandas, Numpy, Matplotlib, Seaborn, Git/Github, Docker, AWS, SLURM, HPC

EXPERIENCE

Computational Biologist

Filbin Lab, Dana Farber Cancer Institute

Boston, MA

Jan 2025 – Present

- Built a reproducible spatial transcriptomics (10x Xenium) analysis pipeline to support multiple projects, improving analysis efficiency and scalability across projects
- Led analysis and optimization of spatial dataset for a Nature-published study, implementing a novel "coherence" metric to quantify tumor organization and reveal recurrence-associated spatial patterns
- Developed custom computational workflows integrating spatial transcriptomics with immunofluorescence (image) and calcium imaging data (video) to uncover celltype-specific molecular and functional patterns

Computational Biology Intern

Novasenta Inc.

Pittsburgh, PA

May 2024 – August 2024

- Identified celltype abundances in Lung Cancer Visium samples using cell2location spot deconvolution, improving company's understanding of localization of target ligand-receptor (L-R) pairs in tumor tissue
- Quantified L-R colocalization in VisiumHD data with binomial test; got significant colocalization, $p < 0.01$
- Implemented Nuclei Segmentation on VisiumHD data to get spatial-single cell data for L-R analysis and found multiple celltype-pair interactions with p -value < 0.05 according to custom-made permutation test

Machine Learning Intern

TenSixty BioSciences, Inc

Cambridge, MA

Sep 2022 – July 2023

- Developed a scalable pipeline for rare cell marker identification in single cell data, including containerized implementation on AWS-parallel for high-throughput isoform-level alignment
- Identified a differentially expressed isoform in small cell lung cancer for targeted drug development and designed a high-accuracy protein binder using RFDiffusion

PROJECTS

Developing a bot for slither.io game using Reinforcement Learning

Self Project

Boston, MA

January 2025 – Present

- Developing a reinforcement learning bot for slither.io game, using PyTorch as framework
- Studied core deep learning and reinforcement learning concepts in depth through Dive into Deep Learning book to build the theoretical background needed for the project

PUBLICATIONS

- Single-cell multidimensional profiling of tumor cell heterogeneity in supratentorial ependymomas. **Nature (2025)**
- The fourth annual Carnegie Mellon Libraries hackathon for biomedical data management, knowledge graphs, and deep learning. **BioRxiv (2023)**